

Matrix functions and stability

Degree coverage with seven matrix multiplications

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Rating rationale: Historical ratings for the original conjecture: challenging because numerical evidence for coefficient-space coverage needs exact algebraic justification; community impact comes from multiplication budgets for practical matrix-function evaluation.

Lean proof and verification evidence

Lean verified — 19 September 2026 (UTC). **Formalization:** George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, with substantial AI assistance. Marcus Webb, The University of Manchester retains mathematical construction authorship.

Seven chronological products cover every complex polynomial through degree 44 in the full 129-coefficient closure model, refuting the original equality 42. The later exact maximum 47 is outside this formalization.

All 25 frozen contracts passed actual local serial Lean, two wholly nonauthor final AI-agent source reviews, and the [sgstepaniants run 35433896875](#) and [ajt60gaibb run 35433898434](#). The real Comparator, default kernel, standard-axiom checks, sandbox and rejection controls passed on the immutable proof [c89c859a](#). Two independent statement reviews preceded implementation. Kernel-mode LeanCert audits are retained, with exact computation minimized. An independent operational audit authenticates the execution evidence. No official Tau Ceti service or external human peer review is claimed.

[Proof, reproduction and scope](#) · [Exact execution evidence](#) · [Formalization metadata](#) · [Upstream PR #305](#).

Resolution — exact maximum 47, 17 September 2026

Author: Marcus Webb, The University of Manchester. Developed with substantial ChatGPT/Codex assistance. Separate Codex agents reviewed the mathematics, and an independently implemented exact verifier reconstructed the full polynomial and its derivatives. This is informal AI-agent review, not external human peer review or formal verification.

[Theorem 1 of the complete proof](#) determines the exact maximum in the retained question:

$$d_7 = 47.$$

Every complex polynomial of degree at most 47 belongs to the Zariski closure of seven-product outputs in the full coefficient space through degree 128. An explicit seven-product family has a rigorously certified order-47 contact with an invertible 48-by-48 coefficient Jacobian. Exact Gaussian-rational contraction bounds prove the contact exists; the inverse function theorem and a weighted input/output limit give the universal coverage statement, with every higher coefficient tending to zero.

A self-contained available-space dimension count gives $\dim \overline{\mathcal{P}_7}^{\mathbb{Z}} \leq 49$. Irreducibility and the seven-product polynomial x^{128} then exclude coverage of the entire degree-at-most-48 space. The new proof does not depend on the earlier border construction or on an external dimension theorem. Exact representation of every degree-47 polynomial and numerically stable coefficient recovery are not asserted.

[Standalone proof source](#) · [Exact certificates and reproduction](#) · [Independent review](#). The original equality $d_7 = 42$, permanent ID and canonical path remain unchanged. The earlier degree-42 and degree-44 results below retain their attribution and archived evidence.

Earlier negative resolution — degree 44, 17 September 2026

Author: Marcus Webb, The University of Manchester. Developed with substantial ChatGPT assistance; two separate Codex agents independently reviewed the proof and recomputed its exact certificates. This is informal AI-agent review, not external human peer review or formal verification.

The conjectured equality is false. [Theorem 1 of the complete proof](#) establishes

$$\mathbb{C}[x]_{\leq 44} \subseteq \overline{\mathcal{P}_7}^Z.$$

The construction first retains a suitable quadruple of polynomials in the joint closure of four-product computations (Lemma 2). Three more products give a 45-parameter coefficient map whose exact Jacobian determinant is 256. The argument uses the original complex field and all coefficients through degree 128; high-degree coefficients vanish in the limit rather than being discarded.

Corollary 3, using Jarlebring–Lorentzon’s dimension theorem, gives $44 \leq d_7 \leq 47$ for the maximum in the retained question. That contribution left the exact maximum undetermined while giving a complete negative answer to the specific equality $d_7 = 42$; the new theorem above now determines it as 47. No exact representation of every degree-44 polynomial, real Euclidean density, or numerical stability is asserted.

[Proof source](#) · [Independent reviews and reproduction](#) · [Source conversation and attribution](#). The permanent ID, canonical path and original mathematical target are retained. The degree-42 result below retains its separate credit.

Verified partial result — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review: PASS for the stated partial results.**

A fixed seven-product scheme has a full-rank complex coefficient map, certified by a nonzero exact integer Jacobian minor. Its image contains a nonempty Zariski-open subset of $\mathbb{C}[x]_{\leq 42}$ and is Euclidean dense there.

This earlier result did not prove an upper bound excluding degree 43 and above, so at the time it left the maximal-degree equality open. It asserted neither exact representation of every polynomial nor real Euclidean dense coverage. The later degree-44 construction above now refutes that equality; the degree-42 theorem remains valid.

Primary reference: [complete authored PDF](#), [standalone TeX](#), [Theorem 1 and equations \(1\)-\(2\)](#). [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Problem statement

Start with the scalar polynomials $1, x \in \mathbb{C}[x]$. Allow arbitrarily many complex linear combinations of already computed polynomials and at most seven multiplications of two already computed polynomials. Let $\mathcal{P}_7 \subseteq \mathbb{C}[x]_{\leq 128}$ be the set of possible outputs.

Identify a polynomial with its coefficient vector in $\mathbb{C}^{129}$ and let $\overline{\mathcal{P}_7}^Z$ be its Zariski closure: the common zero set of all polynomial equations in the coefficients that vanish throughout $\mathcal{P}_7$. Is

$$\max\{d \in \{0, \dots, 128\} : \mathbb{C}[x]_{\leq d} \subseteq \overline{\mathcal{P}_7}^Z\} = 42?$$

The target concerns closure and the complex coefficient field. Exact representability of every polynomial is a stronger assertion.

Why it matters

For a dense input matrix, matrix products dominate the cost of polynomial evaluation. The conjecture determines how much polynomial degree a budget of seven products can cover, providing a theoretical benchmark for matrix-function evaluation schemes.

References

1. E. Jarlebring and G. Lorentzon, *The polynomial set associated with a fixed number of matrix-matrix multiplications*, arXiv:2504.01500v3 (13 August 2025), §2.2 (closure convention), §5.4, Conjecture 13. [Primary text](#).
2. J. Sastre, J. Ibáñez, J. M. Alonso, and E. Defez, *Beyond Paterson–Stockmeyer: Advancing matrix polynomial computation*, WSEAS Transactions on Mathematics 24 (2025), 684–693, §4 (a five-product construction). [Primary paper](#), [DOI](#).
3. J. M. Alonso, J. Sastre, J. Ibáñez, and E. Defez, *A systematic framework for stable and cost-efficient matrix polynomial evaluation*, arXiv:2603.23143v1 (24 March 2026), abstract and conclusions. [Primary text](#).

Status check — 2026-09-08

The latest arXiv source remains v3 and labels the displayed equality a conjecture supported by computations. The later papers give constructions and stability procedures without establishing the seven-product degree-coverage claim. Searches included the exact title with 2026, *Jarlebring Lorentzon conjecture seven multiplications 42*, and *Sastre polynomial evaluation 2026*. No resolution was located. A [3 September 2026 author talk](#) also describes higher-cost degree coverage as conjectural. Six-product real/complex formulations have a separate source ambiguity and are deliberately not included here.

Audit — 2026-09-10

Rechecked [Conjecture 13](#) and [Remark 14](#): complex closure coverage at seven products is still conjectural, with concrete matrix-exponential applications. Title and seven-product searches found no exact proof. The [June 2026 degree-eight construction](#) solves a different budget/degree problem.